Assessment 1

Question 2(a)

- The state can be represented as Python tuple:
  (jug4, jug3)

  jug4 represents the amount of water in the 4-litre jug
  jug3 represents the amount of water in the 3-litre jug

  For example – the state (4,1) represents a situation where the 4-litre jug contains 4 litres of water and 3-litre jug contains 1 litre of water.

  A tuple is an appropriate Python object because it is simple, immutable and can easily be stored in a visited list during a search algorithm.

- Th initial state is (0,0) since initially both jugs are empty.

- The goal states are (2,0), (2,1), (2,2), (2,3) because each of these states contains exactly 2 litres in the 4-litre jug and the goal is to obtain exactly 2 litres of water in the 4-litre jug.

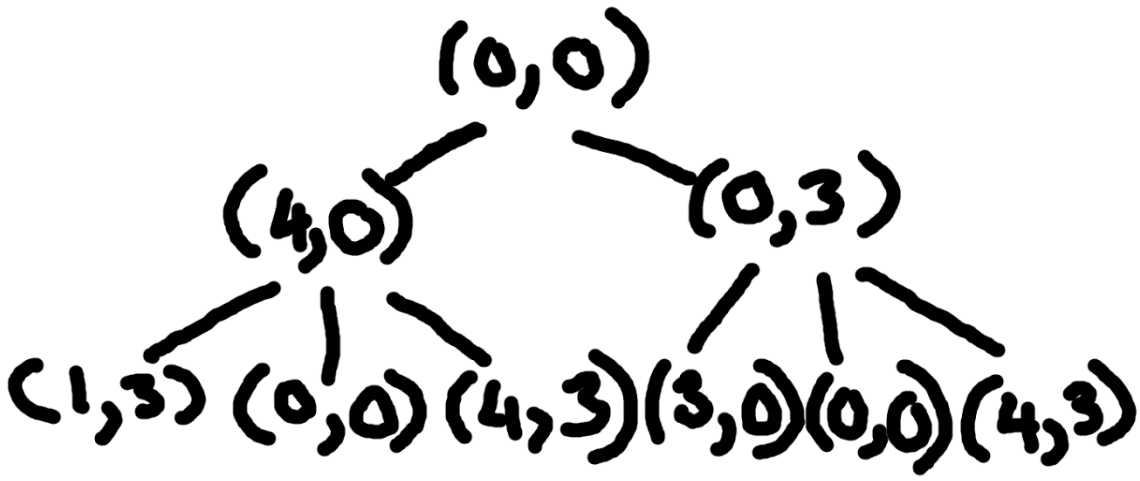
- There are six possible actions:

  1. Fill the 4-litre jug
  2. Fill the 3-litre jug
  3. Empty the 4-litre jug
  4. Empty the 3-litre jug
  5. Transfer water from the 4-litre jug to the 3-litre jug
  6. Transfer water from the 3-litre jug to the 4-litre jug

Question 2(b)

- The 4-litre jug can contain 0, 1, 2, 3, 4 litres of water. Therefore there are 5 possible values.
- The 3-litre jug can contain 0, 1, 2, 3 litres of water. Therefore, there are 4 possible values.
- The total number of states is:
  5 * 4 = 20 possible states

Question 2(c)



From (4,0):

- Transfer 4L - 3L – (1,3)
- Empty 4L – (0,0)
- Fill 3L – (4,3)

From (0,3):

- Transfer 3L – 4L – (3,0)
- Empty 3L – (0,0)
- Fill 4L – (4,3)

- The initial state is (0,0), where both jugs are empty. After one action, the reachable states are (4,0) and (0,3) by filling either jug. Applying one further action from each of these states produces the second level of the search space, giving the states (1,3), (4,3), (3,0) and (0,0). Therefore, there are six unique states that can be reached in two actions or fewer.

Question 2(d)

Action priority:

1. Transfer water between jugs
2. Empty a jug

3. Fill a jug

The goal is to obtain exactly 2 litres in the 4-litre jug.

Step	Stack	Current_node	Visited
1	[(0,0)]	(0,0)	[]
2	[(0,3), (4,0)]	(0,0)	[(0,0)]
3	[(0,3)]	(4,0)	[(0,0), (4,0)]
4	[(0,3), (1,3)]	(4,0)	[(0,0), (4,0)]
5	[(0,3)]	(1,3)	[(0,0), (4,0), (1,3)]
6	[(0,3), (1,0)]	(1,3)	[(0,0), (4,0), (1,3)]
7	[(0,3)]	(1,0)	[(0,0), (4,0), (1,3), (1,0)]
8	[(0,3), (0,1)]	(1,0)	[(0,0), (4,0), (1,3), (1,0)]
9	[0,3)]	(0,1)	[(0,0), (4,0), (1,3), (1,0), (0,1)]
10	[(0,3), (4,1)]	(0,1)	[(0,0), (4,0), (1,3), (1,0), (0,1)]
11	[(0,3)]	(4,1)	[(0,0), (4,0), (1,3), (1,0), (0,1), (4,1)]
12	[(0,3), (2,3)]	(4,1)	[(0,0), (4,0), (1,3), (1,0), (0,1), (4,1)]
13	[]	(2,3)	Goal Found

Question 3(a)

- P(L|D) represents the probability that a person tests positive on a Lateral Flow Test if they have Covid-19.

Question 3(b)

- P(D) = 1/40 = 0.025
- P(¬D) = 1 – 0.025 = 0.975
- P(L|D) = 0.72
- P(¬L|¬D) = 0.999
- P(L|¬D) = 1 – 0.999 = 0.001

Usubg the Law of Total Probability:

- P(L) = P(L|D)P(D) + P(L|¬D)P(¬D)
- P(L) = 0.018 + 0.000975
- P(L) = 0.018975

Question 3(c)

- $P(D|L) = \frac{P(L|D)P(D)}{P(L)}$
- $P(D|L) = \frac{(0.72)(0.025)}{0.018975}$
- $P(D|L) = \frac{0.018}{0.018975}$
- $P(D|L) = 0.9486$

Interpretation: If an individual receives a positive Lateral Flow Test result, there is approximately a 94.86% probability that the individual actually has Covid-19.